

Using Scene Builder to Develop JavaFX Apps

This article is a continuation of a previous one called “Developing a basic GUI application using JavaFX in Eclipse” by the same author. We recommend that those unfamiliar with JavaFX read the earlier article carried in the June 2017 issue of *Open Source For You*.



Scene Builder is a GUI based standalone tool, using which developers can rapidly design and develop JavaFX interfaces with a simple-drag-and-drop operation. Various UI components are available in the library section, which can be dragged and dropped in the working area for quick development and the corresponding code gets generated automatically in the project in your IDE. The best part of using Scene Builder is that you can also make use of stylesheets to load your interface with attractive visual effects.

Scene Builder will automatically create an FXML file for your interface in its background, which is the core of the JavaFX application. FXML is just like XML, which was developed by Oracle, with which we can design the interface of the JavaFX application. It was created to provide better readability of code and to separate the UI components from the application logic. Let's have a look at some of the features of Scene Builder that have made it a favourite in the developer community.

- **Easier UI design:** Controls, containers, shapes, charts, menus, etc, can all be added very easily just by dragging them from the container library. Animations can also be applied effortlessly to complicated interfaces.
- **Editor for FXML:** The FXML file gets created automatically for the designed UI, which can be viewed and edited very easily. Even the FXML files created by other users can be seen and modified.
- **Preview option:** There is a 'Preview' option in Scene Builder, which lets you preview what your interface will look like at any time during the development process. This is very useful since you get to know the actual progress of your UI.
- **Configurable with any IDE:** Scene Builder has been developed to get easily configured with any IDE. The integration process is also very simple and any changes in the FXML get automatically updated in Scene Builder, and vice versa.
- **Support for CSS:** Visual effects can also be added to the application being developed by applying stylesheets, which can be defined at the control level instead of the application level. This really helps to improve the look and feel of the interface. Configuring the CSS file is as simple as configuring a CSS file in HTML. You only have to create a .css file in your IDE or text editor and point that file to your component in Scene Builder. The other advantage is that you can also preview how your